

Stress Management

Mary Worthen; Elizabeth Cash.

► Author Information and Affiliations

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Introduction

Effective stress management techniques vary and often include behaviors that promote physical health, such as healthy nutrition and exercise, as well as strategies that improve cognitive and emotional functioning. Mindfulness-based approaches to stress reduction have recently generated considerable interest among healthcare and epidemiologic researchers.[1][2][3] The concept of mindfulness, which originated in Buddhist practices, is defined as purposeful, nonjudgmental awareness of one's experiences in the present moment.[4] Structured interventions, such as the Mindfulness-Based Stress Reduction program, provide participants with opportunities to learn breathing meditation, body scan techniques, and gentle, yoga-inspired physical exercises.[5] With practice, individuals learn to process emotions, thoughts, and sensations as they arise. Mindfulness practice may help individuals replace automatic reactions and worries about the future with more adaptive, measured responses and greater awareness of the present moment.[6] Results from studies suggested that regular practice can increase mindfulness and improve the capacity to process emotions, thoughts, and sensations as they arise.[7][8][9][10][11][12] Mindfulness-Based Stress Reduction interventions have been adapted for diverse populations, including individuals with chronic or debilitating health conditions and healthy undergraduate and medical students. Results from randomized controlled trials demonstrated improvements in psychological and physiologic processes relevant to health outcomes and stress management.

Some individuals have a greater innate, or trait, capacity for mindfulness. Individuals with higher trait mindfulness who have not participated in formal mindfulness training tend to report better physical health, fewer physiologic symptoms such as pain, and less use of healthcare resources.[13] Trait mindfulness has been associated with lower levels of anxiety and depression in various medical and nonmedical populations.[14] Trait mindfulness may partly reflect a genetic predisposition. Results from a recent epidemiologic study of adolescent twins showed that trait mindfulness was 32% heritable.[15] Results from the same study showed that environmental factors accounted for 66% of the variance in trait mindfulness, suggesting that mindfulness is also a learnable skill. Results from a Mindfulness-Based Stress Reduction study of university undergraduates showed increases in mindfulness and improved psychological outcomes across participants, although the effects may have been more pronounced among those with higher trait mindfulness at baseline.[16] These findings support the value of mindfulness training, including for individuals with high baseline trait mindfulness.

Issues of Concern

Standard MBSR Programs

Standard Mindfulness-Based Stress Reduction programs have demonstrated potential to reduce physiologic dysregulation, including activation of the hypothalamic-pituitary-adrenal axis and autonomic nervous system, as well as inflammation.[17][18] Results from a study of healthy men showed that those with a greater capacity to focus on the present moment experienced less emotional distress and reduced autonomic reactivity, measured by heart rate, during exposure to hypoxic conditions.[19] Neural mechanisms may underlie the reported associations between

mindfulness practices and central nervous system function. Higher trait mindfulness is positively correlated with activity in the anterior cingulate and prefrontal cortices in healthy adults, whereas results from studies showed reduced activity in both regions among individuals with anxiety and depressive disorders.[20] Higher levels of trait mindfulness are also associated with reduced gray matter volume in the amygdala and caudate in healthy adults and greater gray matter volume in the bilateral gyri of adults with generalized anxiety disorder.[20] Additionally, results from studies demonstrated that mindfulness training increased blood flow in the amygdala and hippocampal regions among patients with breast cancer and increased gray matter concentrations in brain regions involved in the norepinephrine and serotonin systems among healthy adults.[21] [22] Evidence of shared neural circuitry suggests possible mechanisms through which mindfulness-based approaches may benefit individuals experiencing prolonged psychological distress or difficulty managing stress.

Mindfulness-Based Stress Reduction

Mindfulness-Based Stress Reduction was initially developed to reduce suffering among patients with chronic pain by teaching skills that cultivate present-moment, nonjudgmental awareness.[5] Participants learn 3 formal mindfulness techniques over 8 weeks: body scan, gentle yoga, and sitting meditation. The body scan is performed while supine, with attention directed systematically and nonjudgmentally toward different regions of the body to encourage relaxed awareness and acceptance of proprioceptive and interoceptive sensations. Hatha yoga consists of gentle movement and stretching sequences that promote awareness of movement and body position. Sitting meditation helps develop a stable cognitive perspective for observing mental events with openness and acceptance without becoming absorbed in distressing thoughts or feelings. Informal mindfulness practice is also encouraged, during which participants cultivate awareness of present-moment experiences, whether pleasant or unpleasant. The goal is to help participants engage with each experience as it arises and respond skillfully to stressful events rather than reacting habitually. With practice, individuals learn to replace automatic reactions and worries about the future with more adaptive, measured responses and greater present-moment awareness.

Traditional MBSR Interventions

Traditional Mindfulness-Based Stress Reduction interventions can be performed without specialized equipment. Many group facilitators and clinicians keep materials available to modify movement-based exercises and yoga poses for participants with physical limitations. A yoga mat, blocks, and a strap are typically sufficient. Results from studies of group-based Mindfulness-Based Stress Reduction interventions suggested that limiting group size to fewer than 20 participants may promote a cohesive dynamic among participants (see Lehrer, Woolfolk, and Sime, *Principles and Practice of Stress Management*. New York: Guilford).[23] Guided practices, including body scan and sitting meditation, are available through electronic media and downloadable resources. Numerous freely available applications and podcasts also provide mindfulness instruction; guided or unguided timed sitting and supine meditations; and guided practices for use during activities such as exercise or cleaning. Many resources are free or available at modest cost, and some offer free trials. Interested individuals may benefit from exploring several options to identify the resources that best meet their needs. Practitioners may also find that preferred approaches vary according to daily circumstances, stress level, and available practice time. Individuals should avoid listening to guided sessions or podcasts while driving because these resources may increase drowsiness.

Mindfulness-based interventions may be adapted to build on participants' existing strengths and address specific stress-management needs. Results from an intervention involving patients with cancer showed benefits after the Mindfulness-Based Stress Reduction program incorporated shorter meditation assignments, less frequent group meetings, and modifications to physical

movements, including yoga poses. Similar modifications have been considered for other groups of patients with cancer, including replacing the usual mindful raisin-eating exercise with a liquid for individuals who have difficulty eating or swallowing, omitting the full-day silent retreat when it may be overly demanding or inappropriate, modifying physical movements to accommodate injury or pain, and using an individual rather than group format to better coordinate sessions with medical appointments.[24] Modified interventions produced improvements in psychological measures similar to those associated with standard Mindfulness-Based Stress Reduction programs, suggesting that these adaptations may remain effective for increasing mindfulness practices. Adapted programs may also be easier to implement and use among individuals with busy schedules and limited flexibility to add a new routine to daily life.

Clinical Significance

Several authors have offered thoughtful conceptualizations of the mechanisms through which mindfulness may reduce stress and alleviate symptoms of illness.[25] Theoretical models focus on metacognitive factors, explore mindfulness within a widely accepted model of stress and coping, and examine how this practice, which originates in Eastern traditions, integrates with Western psychology.[26][27] Results from other recent studies highlighted the need to examine process variables related to the techniques used, individual meditation practice, social factors, and other aspects of Mindfulness-Based Stress Reduction.[28]

Novice mindfulness practitioners also engage in informal practice while learning to observe their own thoughts and sensations and adopt the perspective of a nonjudgmental observer of their own lives. Attending to one's own experience may create a dynamic cognitive interaction that facilitates the ability to respond to ongoing experiences as though they are occurring for the first time, a perspective typically called beginner's mind. This perspective may interrupt automatic reliance on previously conditioned stress reactions. Paradoxically, positive changes may be especially likely to occur when an individual releases the effort to change or control the process. This perspective lies at the core of empirically validated acceptance-based intervention models. Focusing on the present moment may help disrupt habitual reaction patterns and increase flexibility of response. From a cognitive perspective, viewing current circumstances as novel and unique may increase an individual's ability to generate multiple alternative responses. Mindfulness may also address cognitive-behavioral factors, such as self-focused attention, experiential avoidance, and perceived control.[29][30]

Review Questions

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